UCI University of California, Irvine

Decoding Disease, Designing Interventions

Harnessing AI to Bridge Disease Understanding and Therapeutic Design

ICS Honors Seminar



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Information & Computer Sciences

Department of Computer Science

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The Data-to-Intervention Gap

Biological Data Are Growing. Treatments Are Still Limited.

The need

1 in 4 adults

experienced a mental illness

1 in 10 people

aged 12+ had a drug use disorder

U.S., 2024

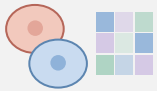
The data

We can measure more than ever



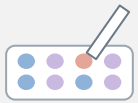
Gene evidence

genetics, literature, and databases



Cell profiles

millions of cells × 20,000 genes



Perturbation data

what happens when genes change

The gap

~90%

of clinical-stage drug candidates fail to reach approval

Data alone don't tell us:

1. **How are cells organized?**
2. **How do cells interact?**
3. **Which genes should we target?**
4. **Which gene combinations could restore a healthy cell state?**

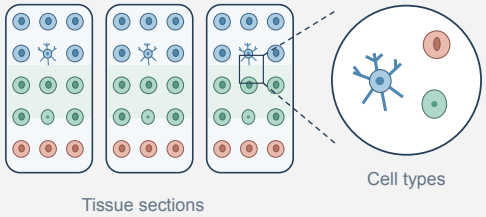
Closing this gap is what our lab works on.

Rounded estimates. 2024 U.S. NSDUH. Drug use disorder: 9.8%, ages 12+. Any mental illness: 23.4%, ages 18+.
~90%: BIO, Informa & QLS, Clinical Development Success Rates 2011–2020 (2021). Phase I to approval: 7.9%.

What Does the Zhang Lab Study?

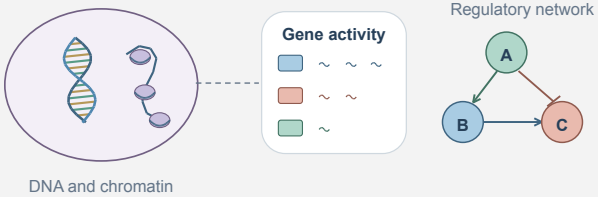
Using AI to understand biological systems and explore interventions

Single-Cell & Spatial Genomics



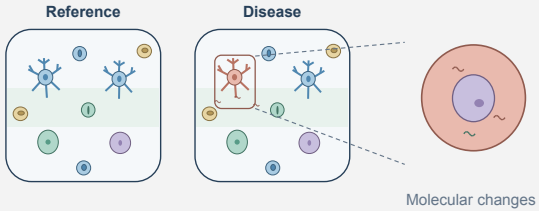
Map cell types and how tissues are organized

Gene Regulation & Functional Genomics



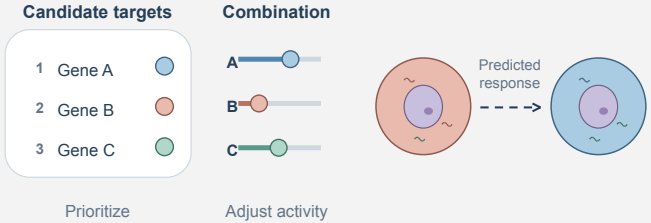
Decode what controls gene activity and cell behavior

Human Disease Genomics



Track how cells change in disease

AI-Guided Therapeutic Discovery

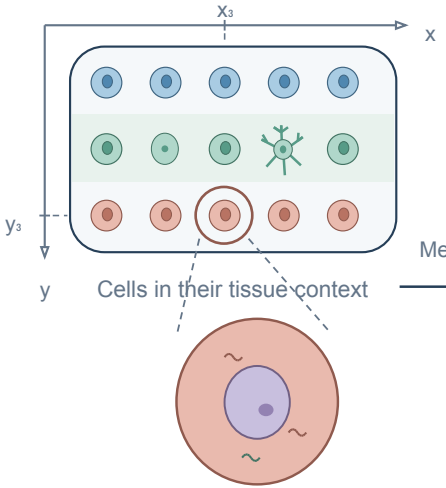


Rank targets to test

From Cells to Data

Measuring gene activity gives each cell a molecular profile

Tissue & cells



Measure RNA for each gene

RNA abundance reflects gene activity

Measure →

Gene expression matrix

Columns = genes

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5	…
Cell 1	9	1	3	0	4	…
Cell 2	7	2	4	1	3	…
Cell 3	1	8	2	6	0	…
Cell 4	2	9	1	7	2	…
Cell 5	0	2	8	3	9	…
Cell 6	1	0	7	2	8	…

Each entry: RNA count for one gene in one cell

Illustrative counts

Cell 3 → [1, 8, 2, 6, 0, …]

Position → (x_3, y_3)

Spatial data: expression + location

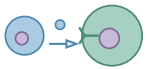
Explore →

Research Questions



MUSE 1

What tissue patterns are shared across slices?



iMiracle 2

How do neighboring cells influence gene activity?



TargetSage 3

Which genes could be therapeutic targets?



CellReprogrammer 4

Which gene combinations shift cells toward a desired state?

01

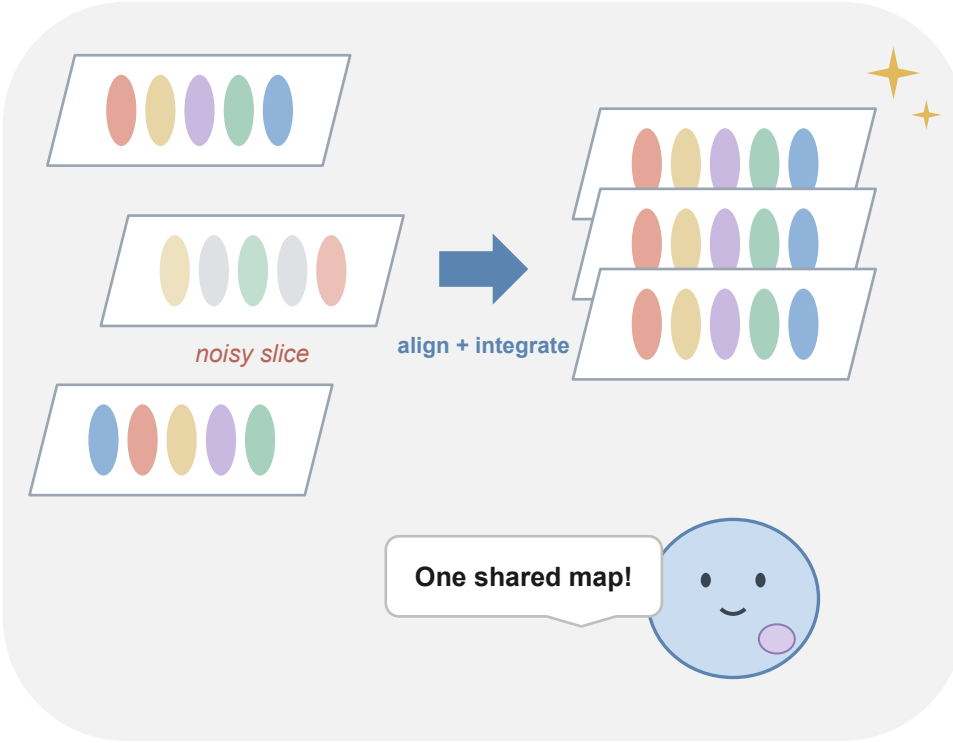
DECODING DISEASE

MUSE

CIKM 2025

What tissue patterns are shared across slices?

Adjacent slices share structure, so cleaner slices can help noisier ones.

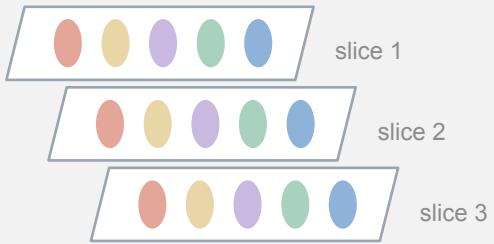


MUSE — iMiracle — TargetSage — CellReprogrammer

Multi-Slice Integration: Alignment and Data Quality

Misalignment and noise obscure shared tissue patterns.

One Tissue, Many Slices



A tissue is cut into thin slices. Each spot has an (x, y) location plus counts for ~20,000 genes.

Slices Don't Match Up



Each slice is measured on its own, so the same tissue region can look different from slice to slice.

Some Slices Are Noisy



Low-quality slices have missing or noisy measurements that hide the real patterns.

MUSE's goal: match spots across slices and let clean slices help noisy ones → one shared tissue map

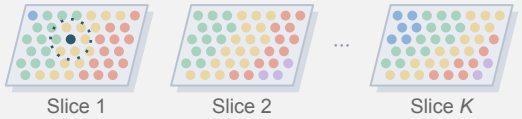
MUSE: Duan et al., CIKM 2025.

MUSE: Cross-Slice Alignment and Quality-Aware Integration

Three steps that upgrade existing single-slice methods to many slices

1 Build Spatial Graphs

Input: multi-slice data

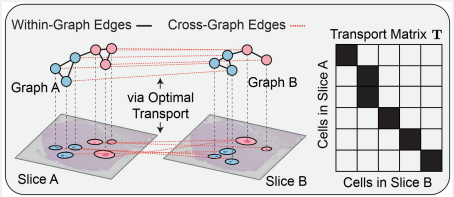


Spatial graph per slice



Link each spot to its nearest neighbors, so every spot knows its local context.

2 Match Across Slices



$$\min_{\mathbf{T}\geq 0} \; (1-\alpha) \underbrace{\Sigma_{i,j'} \delta(\boldsymbol{x}_i^{(\mathrm{A})}, \boldsymbol{x}_{j'}^{(\mathrm{B})})\, T_{ij'}}_{\text{gene expression similarity}}$$

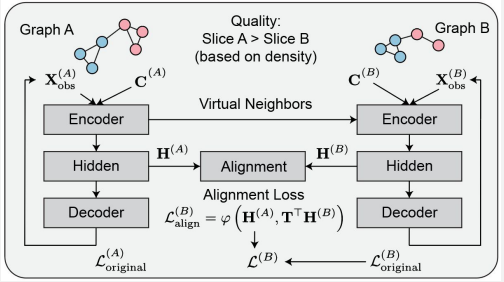
$$+ \alpha \underbrace{\Sigma_{i,p,j',q'} (D_{ip}^{(\mathrm{A})} - D_{j'q'}^{(\mathrm{B})})^2\, T_{ij'}\, T_{pq'}}_{\text{spatial context preservation}}$$

$$\text{s.t. } \mathbf{T1} = \mathbf{a}, \quad \mathbf{T}^{\mathrm{T}}\mathbf{1} = \mathbf{b}$$

Alignment criteria:

- similar expression profiles
- similar local neighborhoods

3 Quality-Aware Fusion



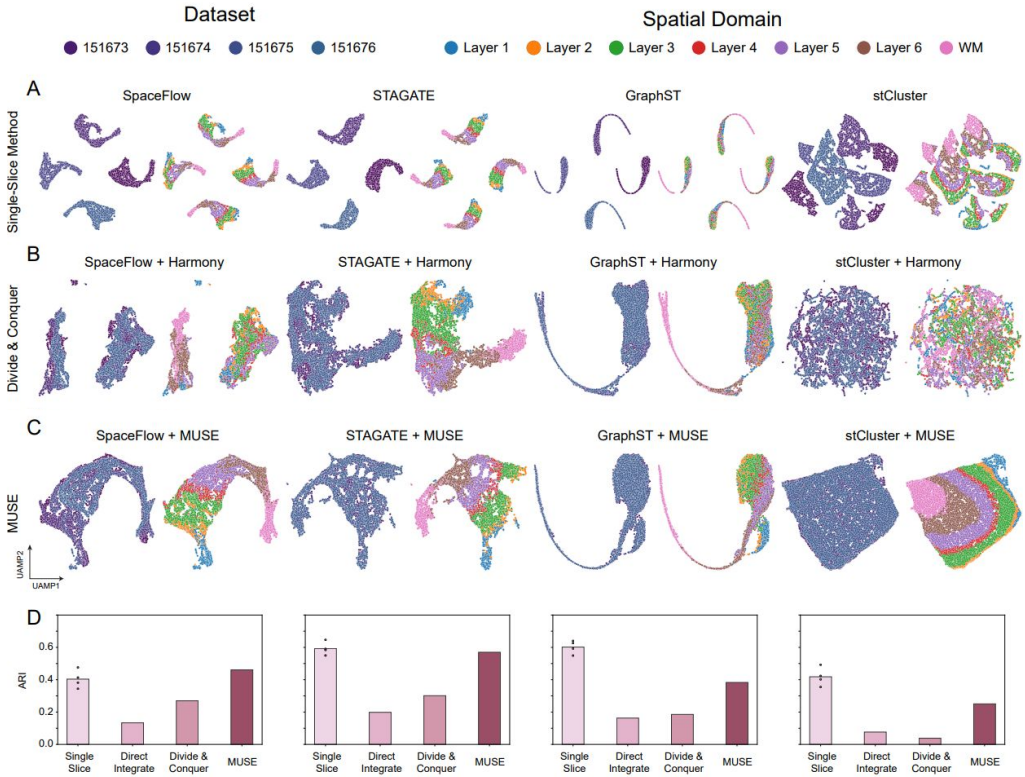
Virtual neighbors:

$$i = \arg\max_i T_{ij'}, \qquad N_{j'} = N_{j'}^{\text{original}} \cup \{i\}$$

Noisy slices borrow “virtual neighbors” from a reference slice.

MUSE: Duan et al., CIKM 2025.

MUSE Learns Coherent Embeddings Across Slices



Human DLPFC (10X Visium)
4 adjacent cortical slices

Single-slice methods:
Fragmented by dataset

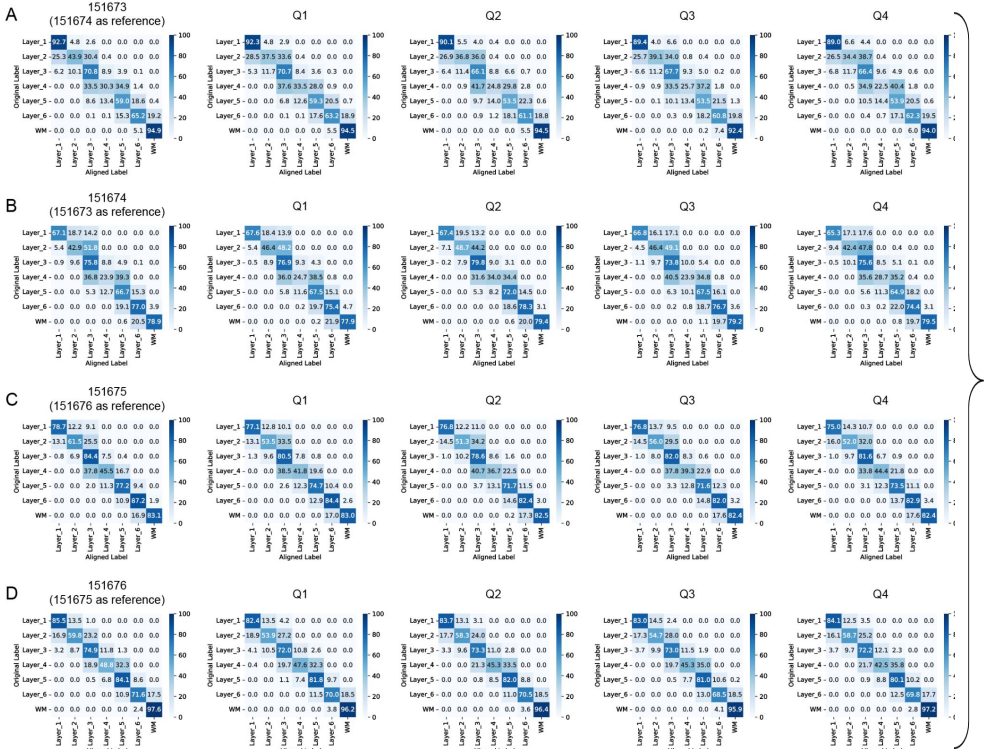
Post-hoc correction:
Slices still form separate groups

MUSE integration:
Aligned across slices

ARI comparison:
Improved domain recovery

MUSE: Duan et al., CIKM 2025.

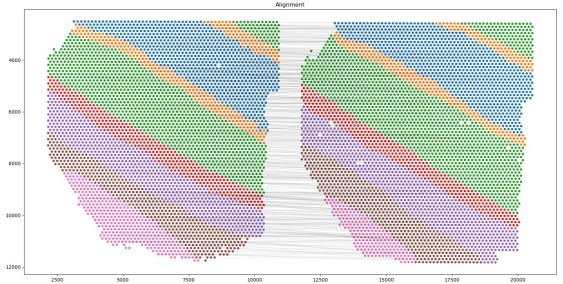
MUSE Aligns Corresponding Tissue Regions Across Slices



Strong diagonal signals
accurate layer-to-layer matching

Robust across quality levels
stable from Q1 to Q4

Consistent across references
reproducible alignment patterns



MUSE accurately matches spots from corresponding cortical layers.

MUSE: Duan et al., CIKM 2025.

02

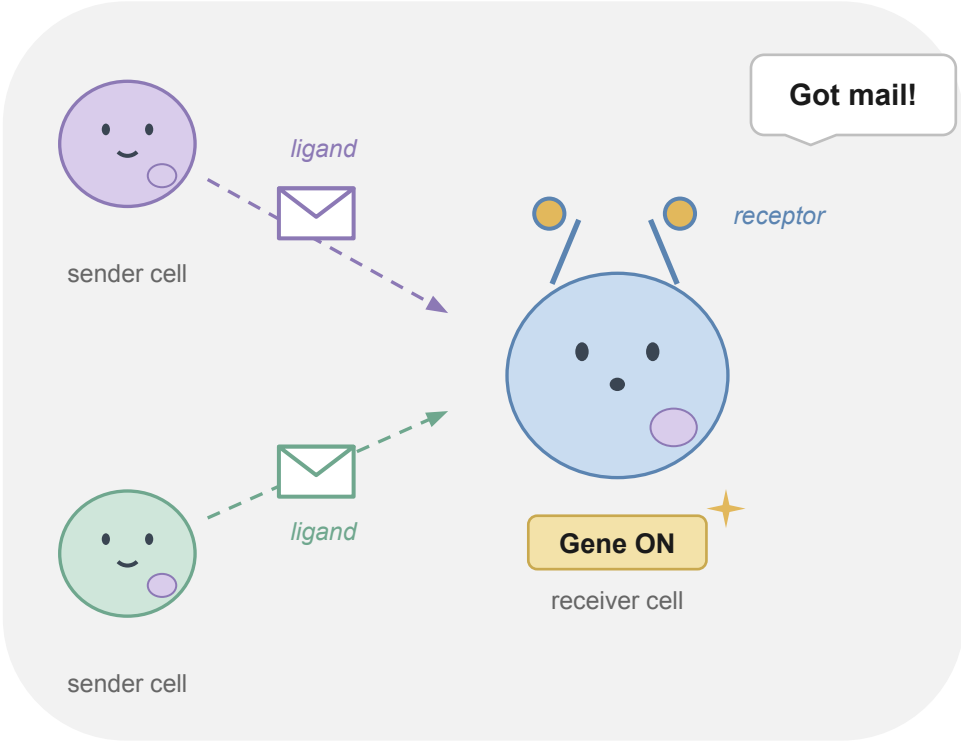
DECODING DISEASE

iMiracle

CIKM 2024

How do neighboring cells influence gene activity?

A cell's gene activity combines its own baseline with signals from its neighbors.

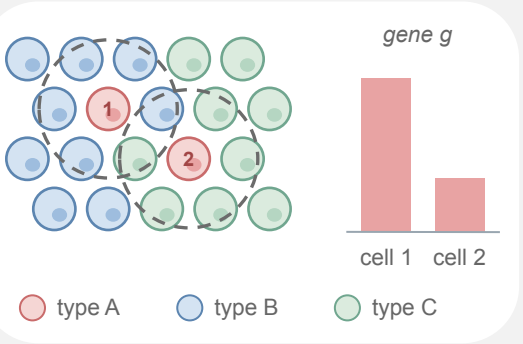


MUSE — **iMiracle** — TargetSage — CellReprogrammer

Linking Cell–Cell Signals to Gene Expression

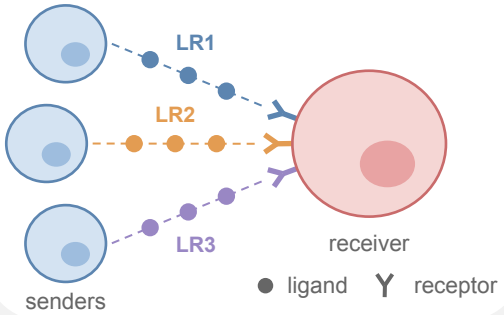
Which signals contribute to each gene's expression in each cell?

Tissue context



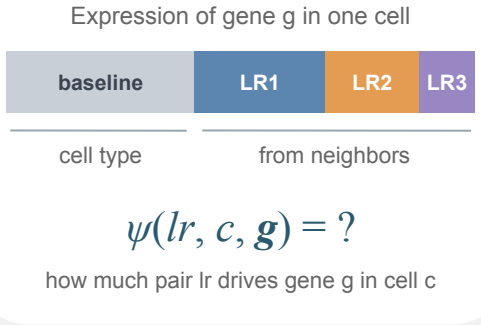
Same cell type, different neighbors → different gene activity.

Cell–cell communication



Ligands from many senders bind receptors on one receiver.

Regulatory interpretation



Which signal explains how much of each gene, in each cell?

Most existing methods: which cells communicate → **iMiracle: what those signals do to each gene, in each cell**

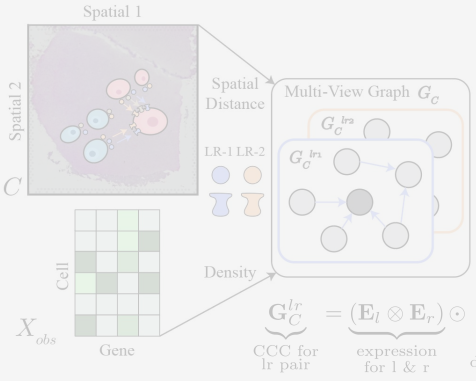
iMiracle: Duan et al., CIKM 2024.

iMiracle: Baseline + Neighbor-Driven Regulation

Multi-View Graph Construction

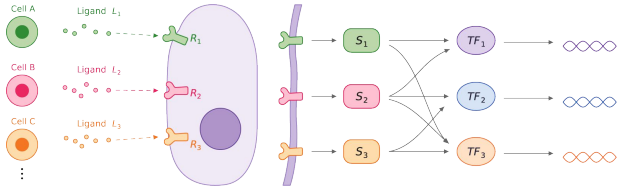
Ligands: Molecules that bind to receptors to trigger a cellular response

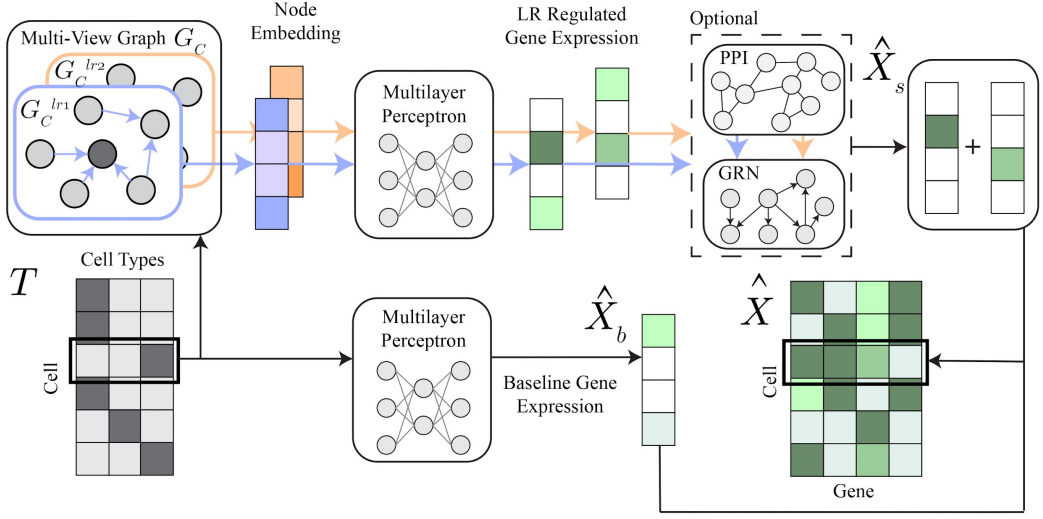
Receptors: Proteins on cell surfaces that receive signals from ligands



$$\underbrace{\hat{\mathbf{X}}}_{\text{gene expression}} = \underbrace{\hat{\mathbf{X}}_b}_{\text{based on cell types}} + \underbrace{\sum \hat{\mathbf{X}}_s^{lr'}}_{\text{based on CCC via lr}}$$

Inter- and *intra*-cellular networks





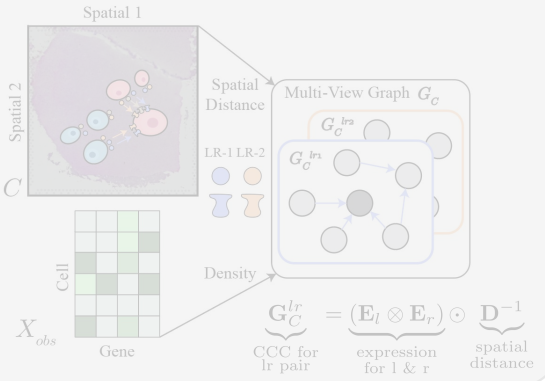
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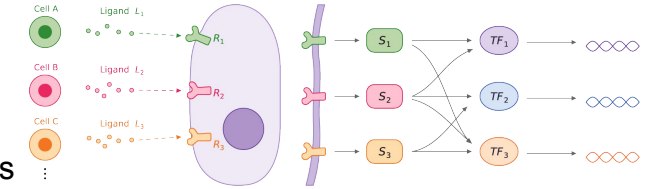
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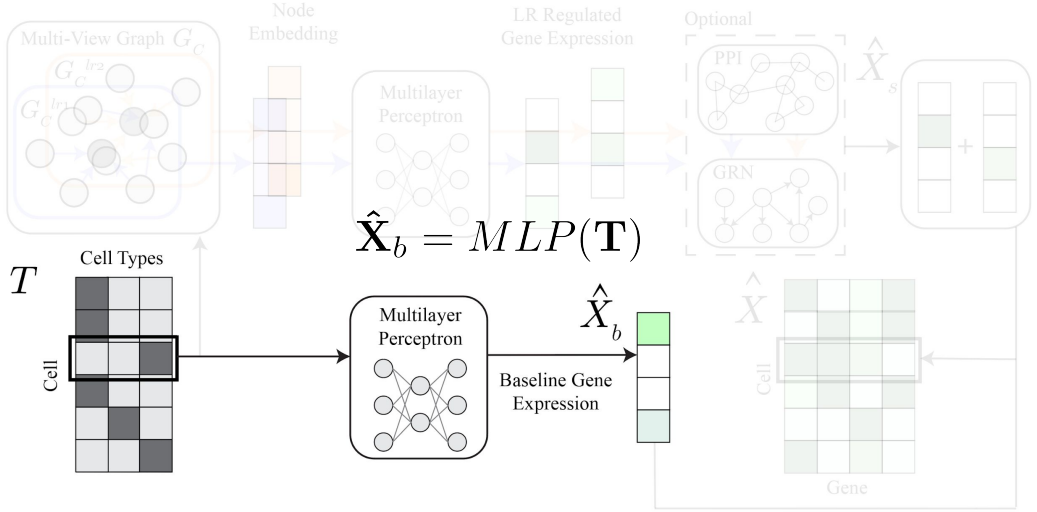
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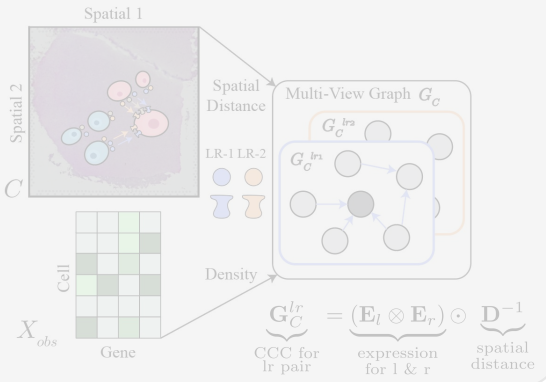
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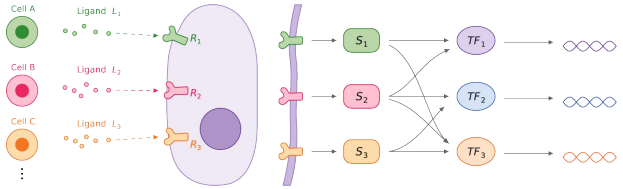
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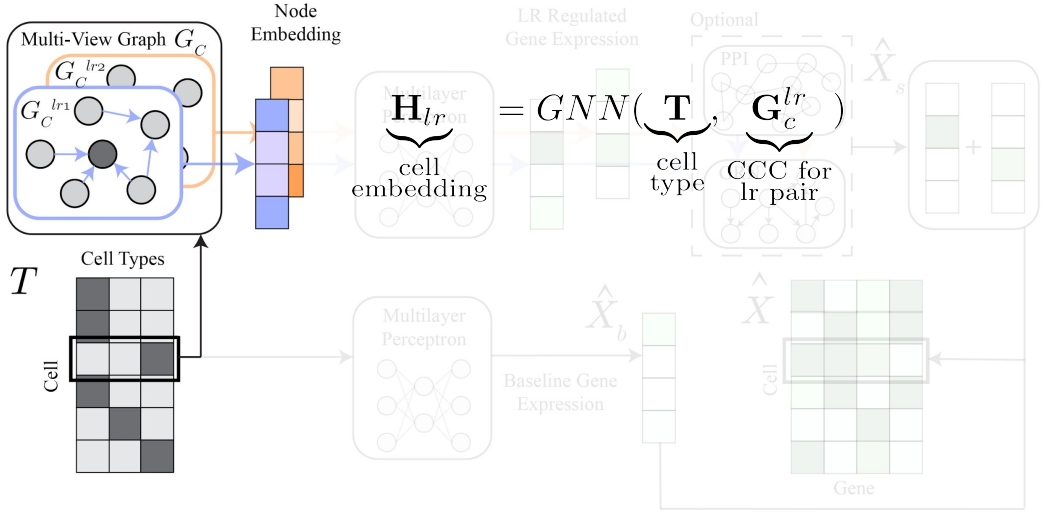
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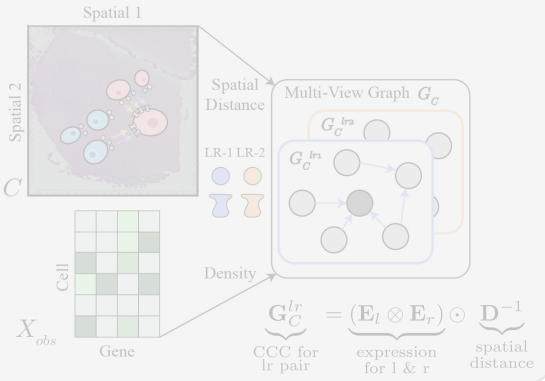
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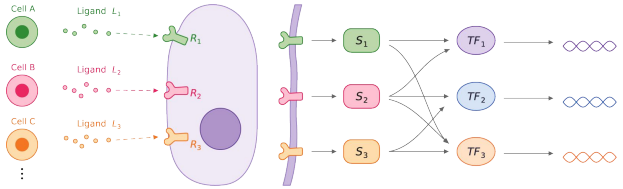
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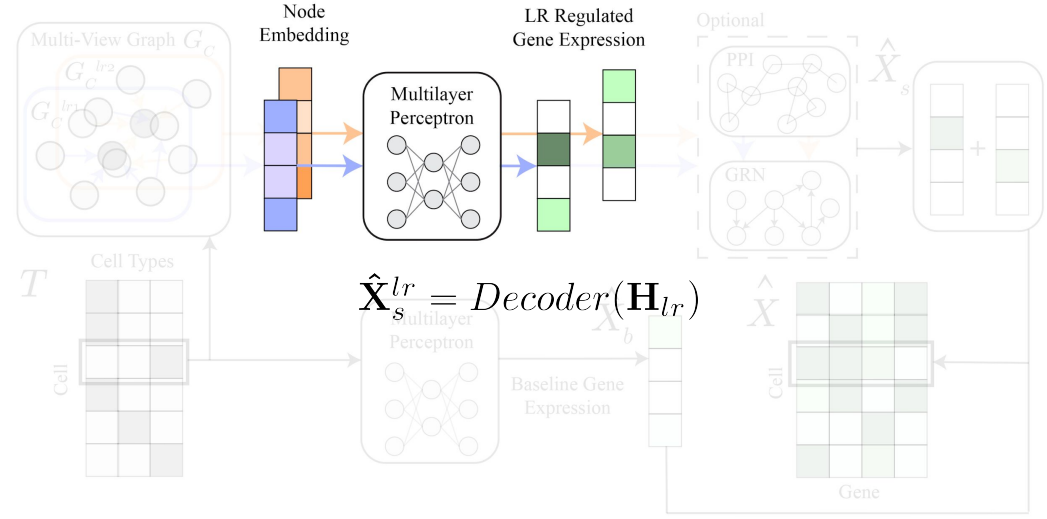
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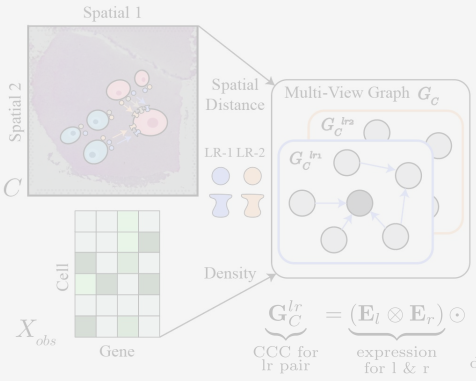
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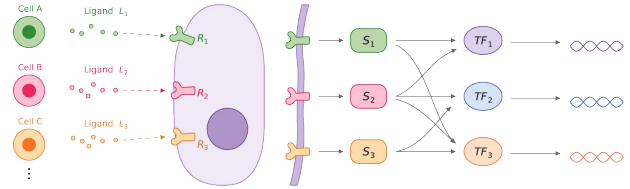
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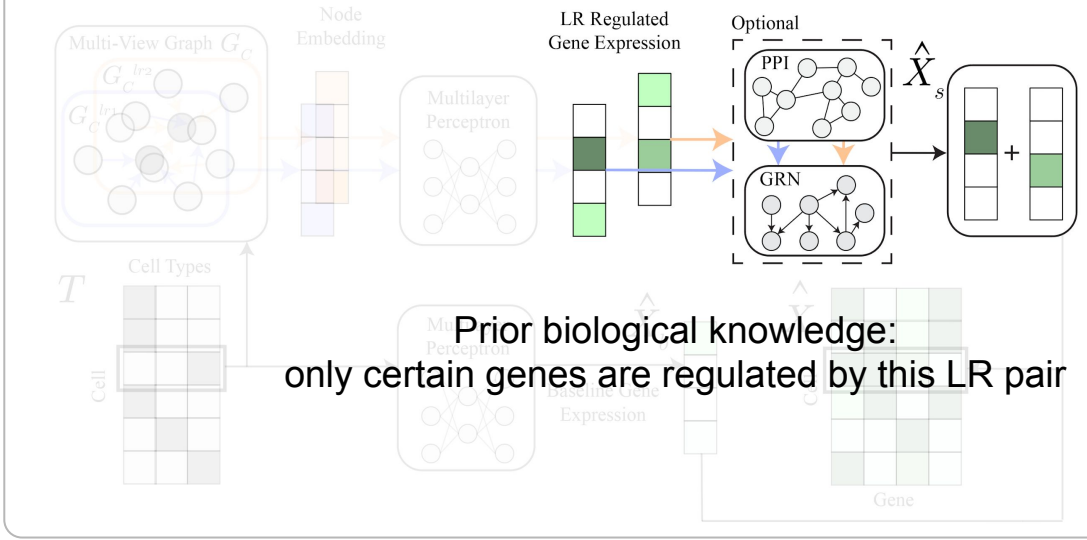
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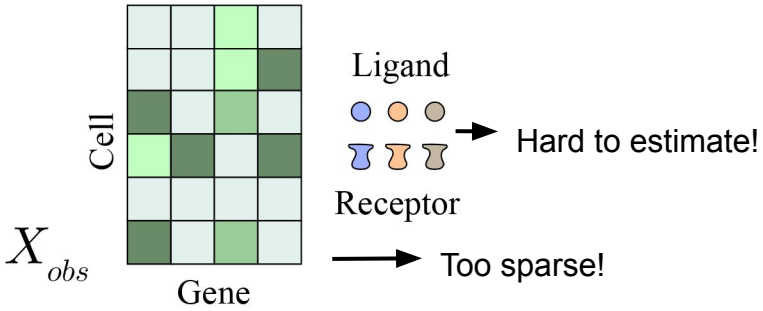
Prior biological knowledge:
only certain genes are regulated by this LR pair

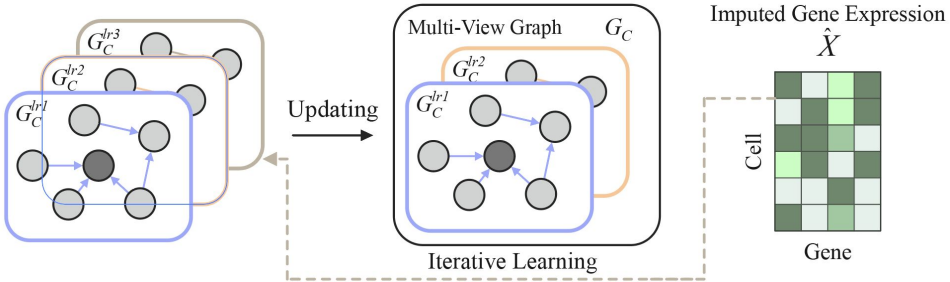
iMiracle: Duan et al., CIKM 2024.

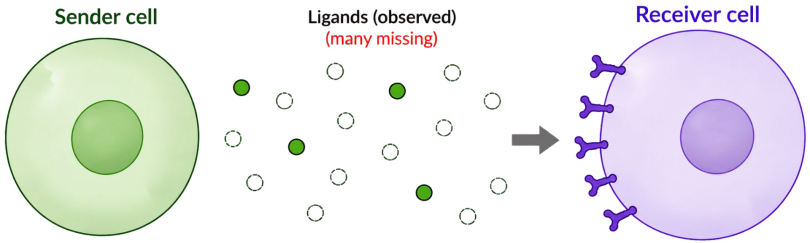
iMiracle: Iterative LR Density Estimation

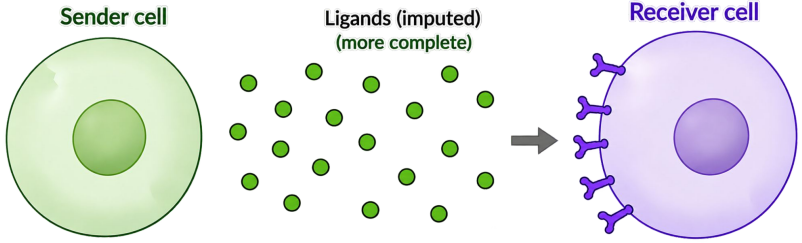
$$\underbrace{\mathbf{G}_C^{lr}}_{\text{CCC for lr pair}} = \underbrace{(\mathbf{E}_l \otimes \mathbf{E}_r)}_{\text{expression for l \& r}} \odot \underbrace{\mathbf{D}^{-1}}_{\text{spatial distance}}$$

$$\mathbf{G}_c^{lr(i+1)} = \alpha \mathbf{G}_c^{lr(i)} + (1-\alpha)\mathbf{G}_c^{lr(i+1)}$$



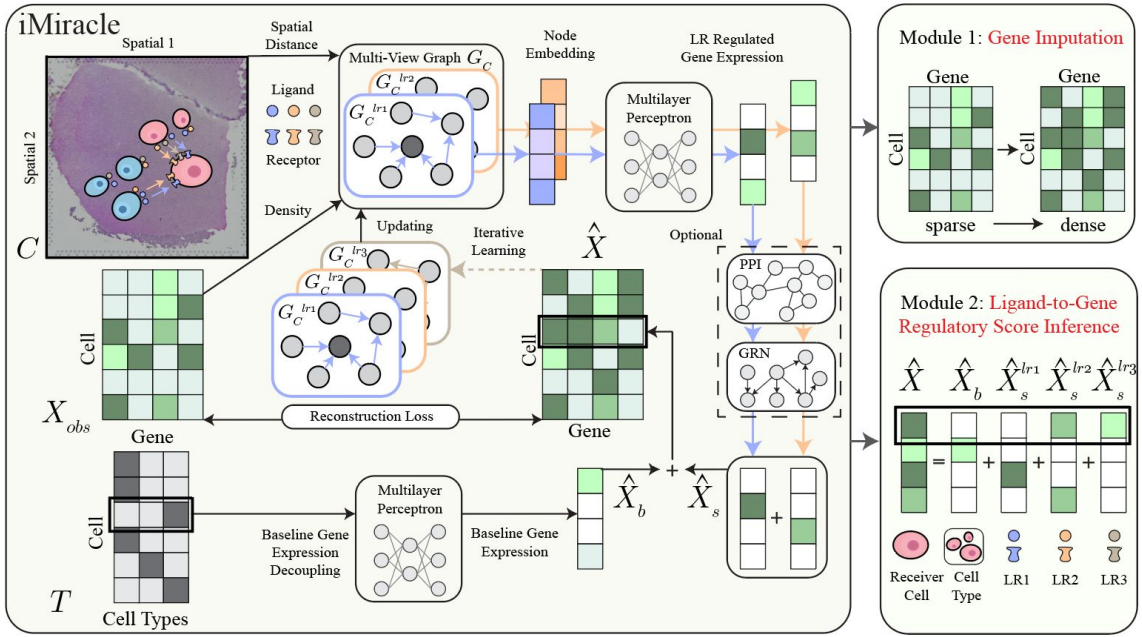






iMiracle: Duan et al., CIKM 2024.

Method Overview



Input

Cell-by-gene matrix

$$\boldsymbol{X}_{obs} \in \mathbb{R}^{n \times m}$$

Cell positions

$$\boldsymbol{C} \in \mathbb{R}^{n \times 2}$$

Cell-by-cell-type matrix

$$\boldsymbol{T} \in \mathbb{R}^{n \times t}$$

Output

Imputed cell-by-gene matrix

$$\hat{\boldsymbol{X}} \in \mathbb{R}^{n \times m}$$

Cell-specific regulatory score

$$\psi(lr, c, \boldsymbol{g}) = \hat{\boldsymbol{X}}_s^{lr'}[c, \boldsymbol{g}] \in \mathbb{R}$$

Model training

$$\mathcal{L} = \frac{\sum_{i,j} \mathbf{1}[X_{obs,(i,j)} \neq 0] \, (\hat{X}_{(i,j)} - X_{obs,(i,j)})^2}{\sum_{i,j} \mathbf{1}[X_{obs,(i,j)} \neq 0]}$$

Reconstruction error over observed (non-zero) entries, so no ground-truth regulation is needed.

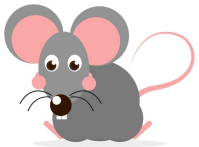
iMiracle: Duan et al., CIKM 2024.

iMiracle Results: CCC Recovery & Gene Imputation

Datasets

2 species
- dorsolateral prefrontal cortex
- mouse olfactory bulb

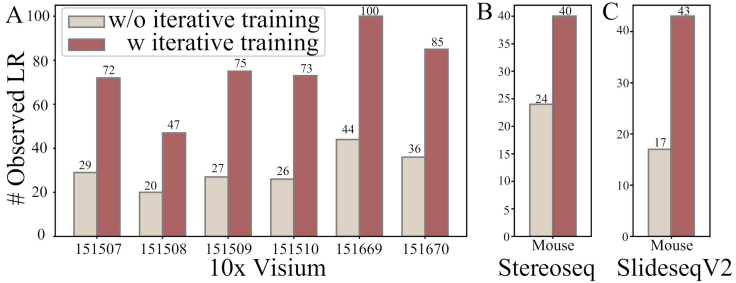




3 sequencing platforms
- 10X Visium
- Stereo-seq
- Slide-seqV2

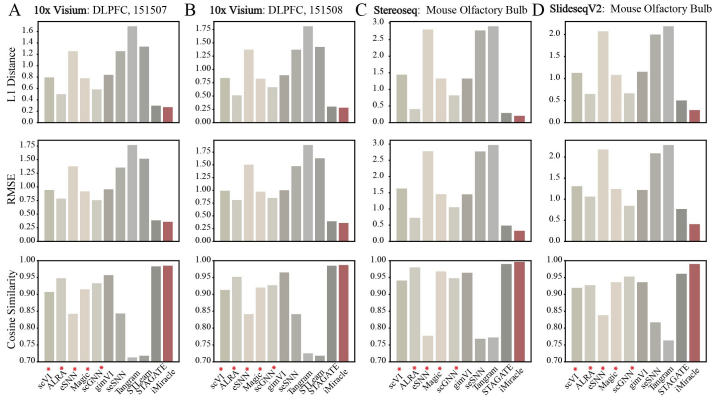
8 datasets

Iterative learning expands the inferred LR network



iMiracle improves the number of detected LR interactions.

iMiracle shows improved gene imputation accuracy

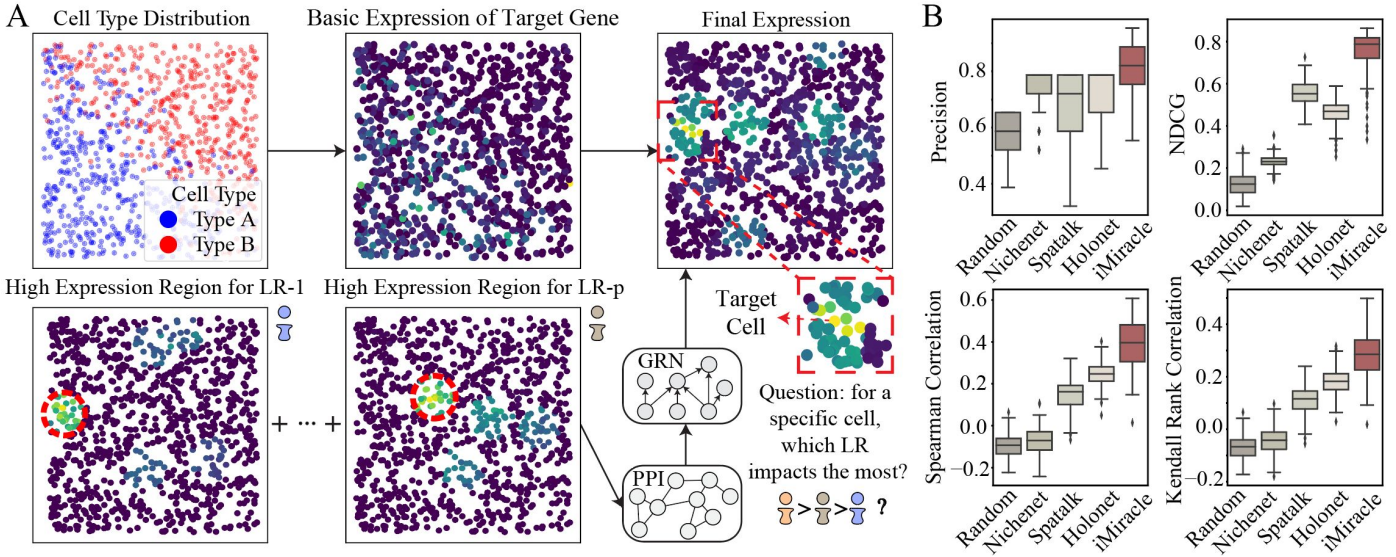


We can confidently rely on imputed gene expression to estimate LR density!

iMiracle: Duan et al., CIKM 2024.

iMiracle Results: LR–Gene Regulation

iMiracle infers accurate inter-cellular ligand-gene regulatory scores



Simulated datasets:

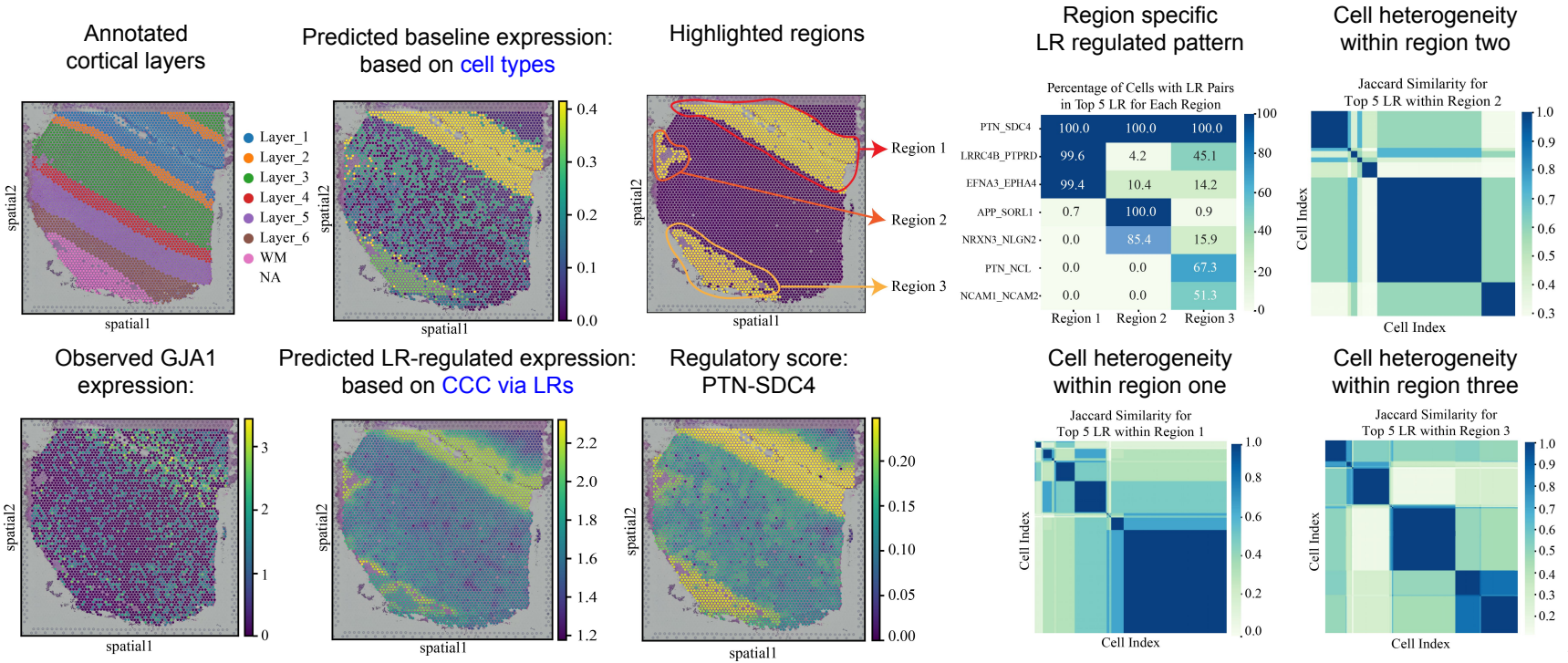
We know the ground truth, specifically which LR contributes the most to a gene in a given cell.

iMiracle is more accurate than random guess and methods with/without spatial information!

iMiracle: Duan et al., CIKM 2024.

iMiracle Results: Regulatory Heterogeneity

iMiracle reveals substantial regulatory heterogeneity across cells



iMiracle: Duan et al., CIKM 2024.

03

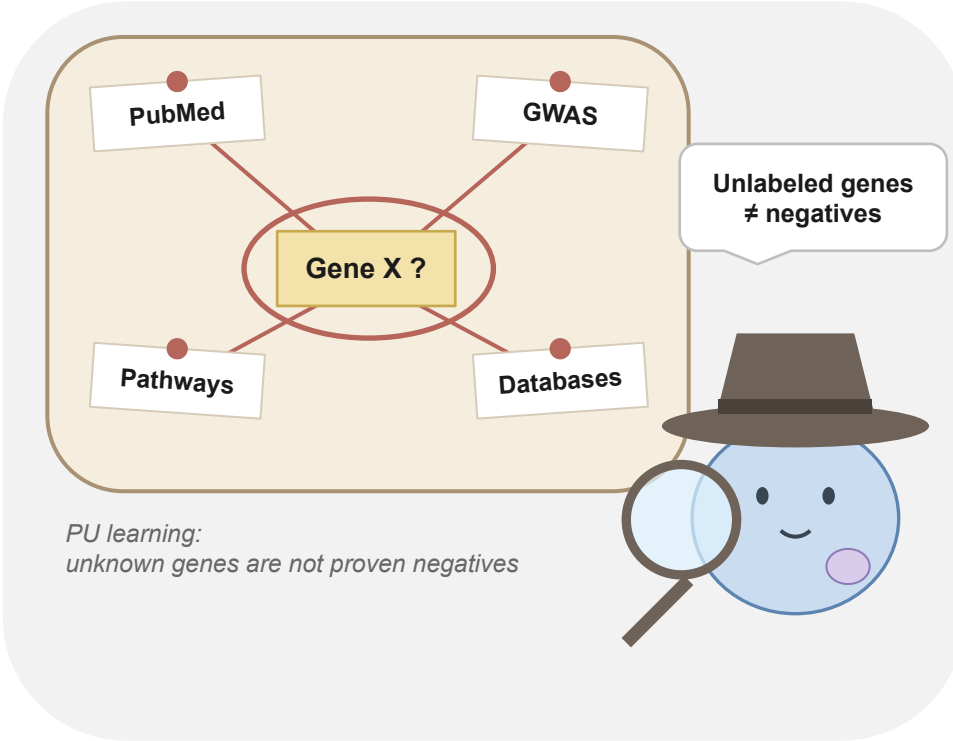
DESIGNING INTERVENTIONS

TargetSage

NeurIPS 2026

Which genes could be therapeutic targets?

Evidence is scattered across sources, and only a few hundred genes are known targets.

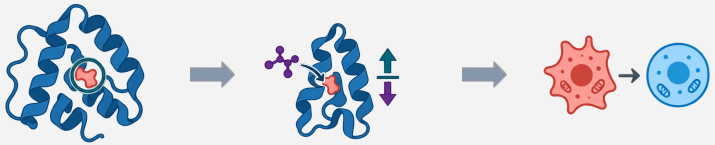


PU learning:
unknown genes are not proven negatives

MUSE — iMiracle — **TargetSage** — CellReprogrammer

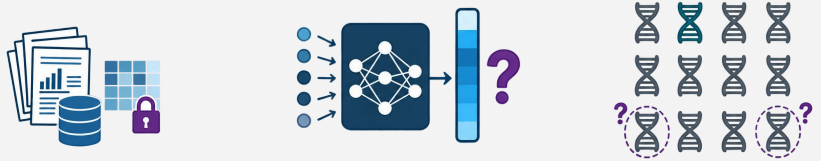
Target Identification: Challenges and Our Approach

What are we looking for?



Identify a target | Modulate activity | Therapeutic benefit

Three challenges



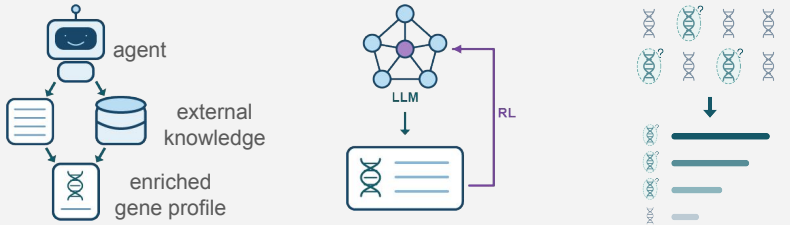
Static features | Limited interpretability | Incomplete labels

How do existing methods rank candidates?



DrugnomeAI: 324 gene-level features from 15 data sources

How does TargetSage address them?



Agentic Profiling | Guided Reasoning | PU Learning

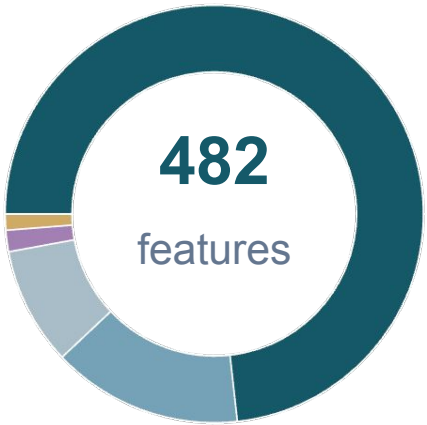
TargetSage: Duan et al., NeurIPS 2026.

DrugnomeAI: Raies et al., Commun. Biol. 5, 1291 (2022). doi:10.1038/s42003-022-04245-4

TargetSage: Integrating Evidence to Prioritize Targets

Biological evidence

19,032 human protein-coding genes



- **Function & networks (353)**
- Chemical & antibody links (71)
- Genomic constraint (44)
- Phenotype & disease (8)
- Expression & sequence (6)

+ Retrieved biomedical evidence (literature & databases)

Training examples

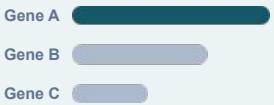
700 known drug targets

18,332 unlabeled genes

PHAROS: a database of drug targets

TargetSage

Target prioritization

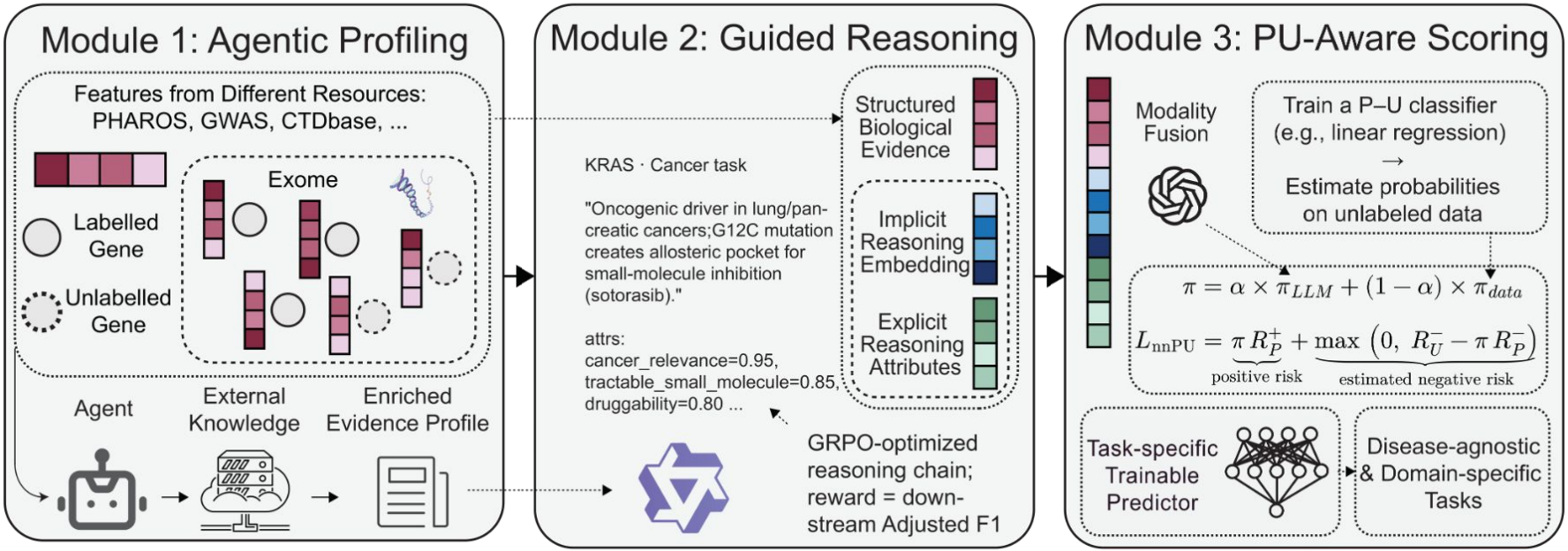


Gene-specific rationales

TargetSage: Duan et al., NeurIPS 2026.

TargetSage Turns Evidence Into Target Scores

INPUT: Gene evidence + known targets → OUTPUT: Target scores + rationales



482 features · 15 databases
6-tool evidence-retrieval agent

23 therapeutic attributes
GRPO-guided gene-specific reasoning

Gene-adaptive multimodal fusion
Hybrid prior + non-negative PU risk

TargetSage: Duan et al., NeurIPS 2026.

TargetSage Leads Across 15 Benchmark Tasks

Adjusted F1 = R_{soft}^2 / $\bar{p}$ ↑

R_{soft}: mean score on known targets

$\bar{p}$: mean score across evaluated genes

○ Best baseline per task

◆ TargetSage

Pharos labels

2 tasks

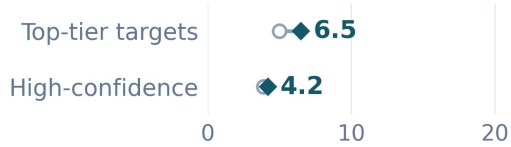
Targets of drugs and active compounds



Druggability tiers

2 tasks

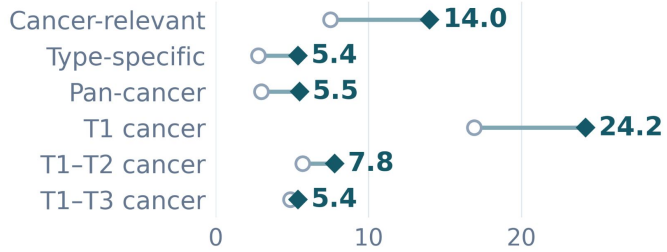
Different levels of supporting evidence



Cancer druggability

6 tasks

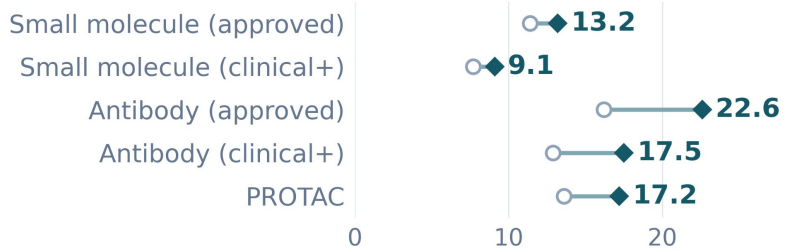
Potential targets for cancer treatment



Drug modality

5 tasks

Small molecules, antibodies and protein degraders



TargetSage: Duan et al., NeurIPS 2026.

PU learning: Bekker & Davis, Mach. Learn. 109, 719–760 (2020). doi:10.1007/s10994-020-05877-5

TargetSage Recovers Held-Out and Clinical-Stage Targets

Temporal label holdout and independent clinical-target enrichment

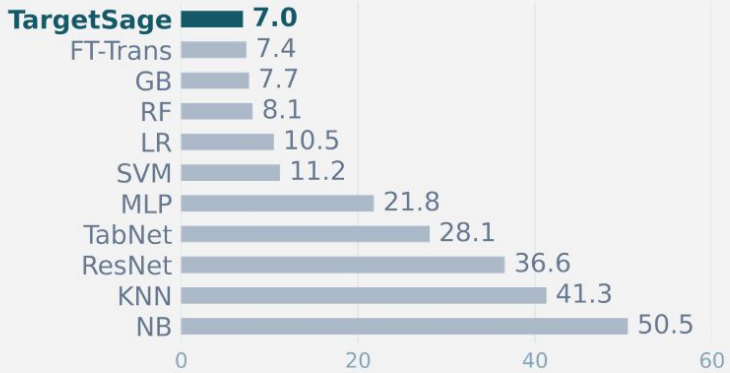
Temporal label holdout

2021 → **2025**

Training labels

90 newly clinical targets

Median rank percentile (%) ↓



Lowest median rank percentile

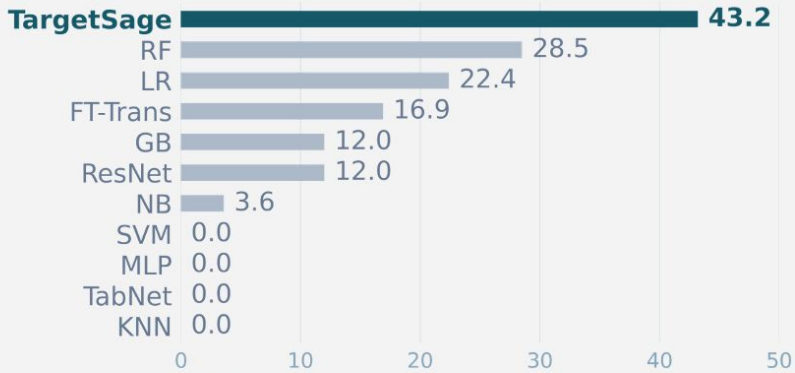
Independent clinical targets

PHAROS 2025 → ChEMBL Phase II/III

136 clinical targets among 11,891 Tbio genes

proteins targeted by drugs in phase II or III clinical trials.

Top-25 enrichment odds ratio ↑



+51% vs. RF (43.2 vs. 28.5)

TargetSage: Duan et al., NeurIPS 2026.

04

DESIGNING INTERVENTIONS

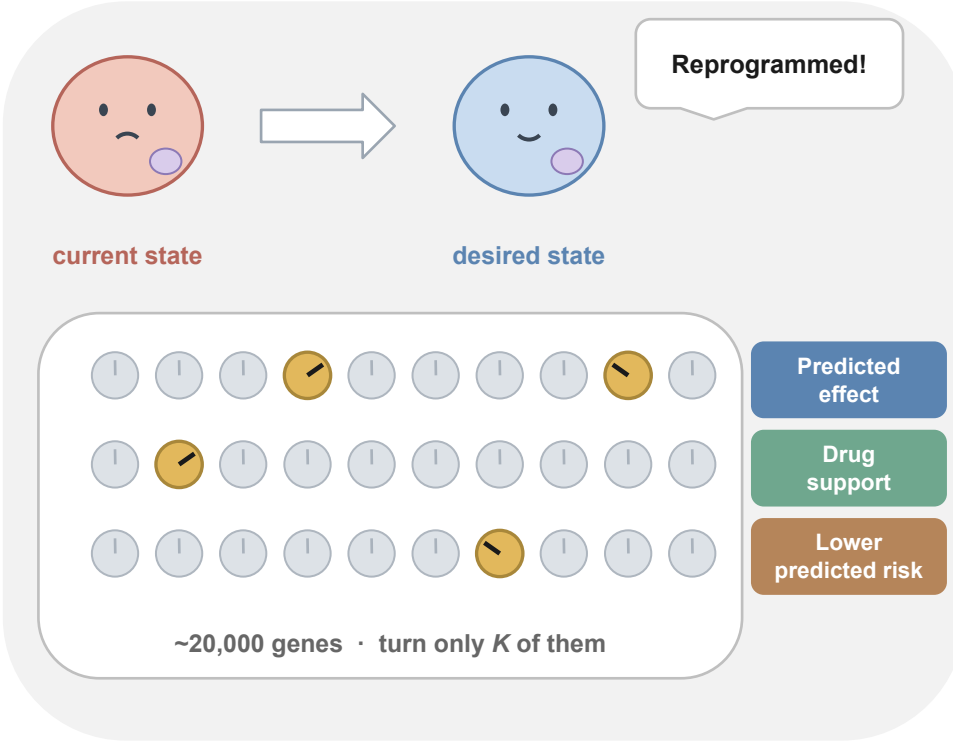
CellReprogrammer

ICLR 2027 · under review

Which gene combinations shift cells toward a desired state?

Genes interact, so the best few must be chosen together, not one at a time.

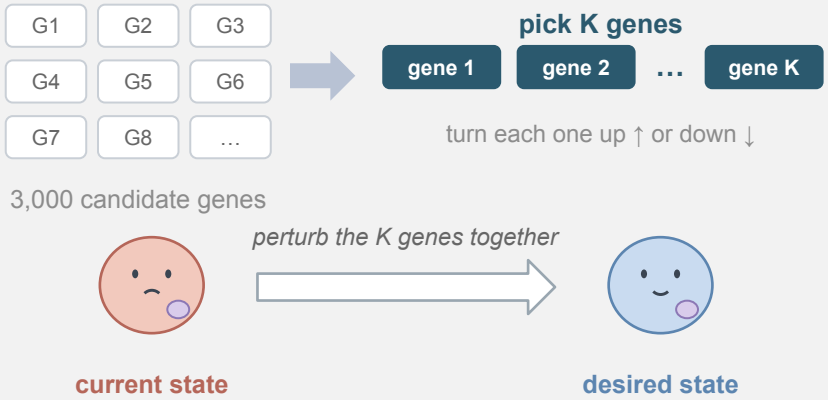
MUSE — iMiracle — TargetSage — **CellReprogrammer**



What Is Cell-State Reprogramming?

Perturb a few genes so that cells move toward a desired state

The task



The desired state depends on the goal:

disease → healthy	healthy → disease	cancer → death
this work		

What makes a good gene set?

Effective
Predicted shift toward a control-like state

Feasible
Only K genes, and existing drugs can move them

Low toxicity
Avoid high-risk targets
and limit excessive expression changes

Hard for computers too: "3,000 choose K" possible gene sets, and the effects of genes do not simply add up.

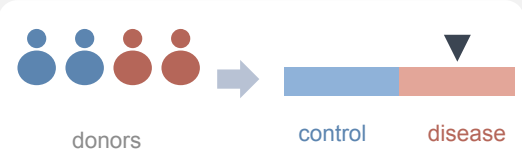
CellReprogrammer: Duan et al., ICLR 2027 (under review).

CellReprogrammer: Optimizing Gene Combinations

Balance predicted effectiveness, druggability, and toxicity.

1 Disease Classifier

How disease-like is this cell state?



Learn a disease score from donor-level expression profiles of patients and controls.

2 Perturbation Surrogate

What if we turn genes up or down?



Fine-tune scGPT on Perturb-seq data to predict gene expression responses to candidate perturbations.

3 Combination Optimizer

Which K genes should we pick?



Update gene scores by gradient descent through the predictor and classifier, using differentiable top-K selection.

Objective

$$\min \; L(\mathrm{S}) + \lambda_{\mathrm{drug}} L_{\mathrm{drug}}(\mathrm{S}) + \lambda_{\mathrm{tox}} L_{\mathrm{tox}}(\mathrm{S}) \quad \text{s.t.} \; |S| = K$$

- $L(\mathrm{S})$: Disease score after perturbation
- $\lambda_{\mathrm{drug}} L_{\mathrm{drug}}(\mathrm{S})$: Penalty for unsupported drug directions
- $\lambda_{\mathrm{tox}} L_{\mathrm{tox}}(\mathrm{S})$: Penalty for essential-gene inhibition
- $|S| = K$: K-gene constraint; bounded expression changes

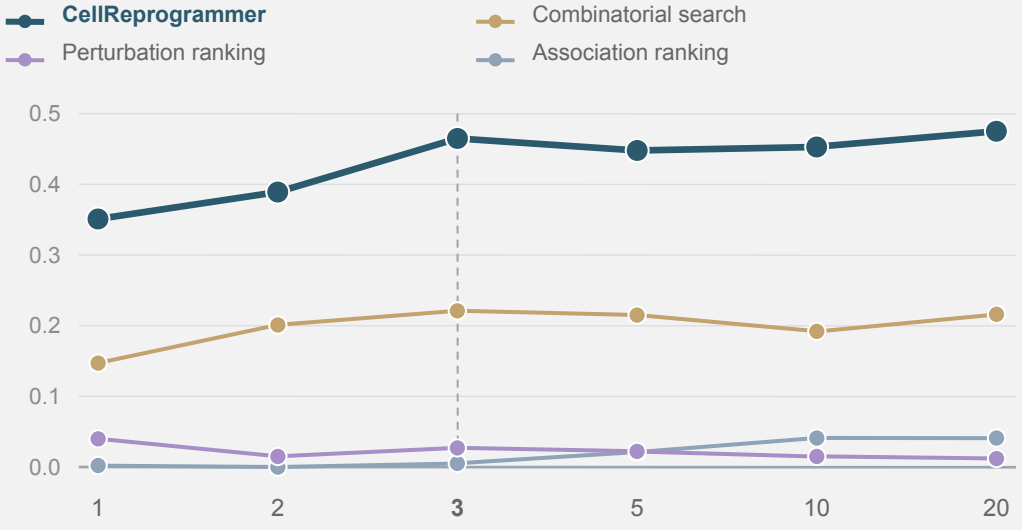
CellReprogrammer: Duan et al., ICLR 2027 (under review).

Strongest Predicted Reprogramming at Every Budget K

5 disorders × 3 cell types · 20 baselines · held-out donors

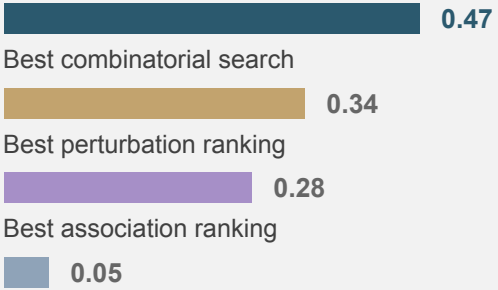
Effect across gene budgets K

Fixed transcriptomic-change cap (κ = 1)



y: predicted effect relative to matched random perturbations · baseline lines = category means

At the default budget K = 3



+38%

vs. the best baseline

15 / 15

settings outperformed matched random perturbations

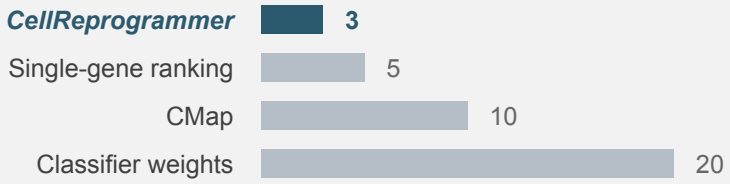
CellReprogrammer: Duan et al., ICLR 2027 (under review).

Compact, Druggable, and Lower Predicted Toxicity

5 disorders × 3 cell types

FEASIBILITY

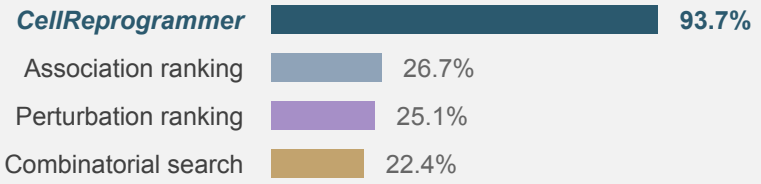
Genes needed for the same effect



Size-dependent cap (κ = K/3) · to reach 0.30 · 10 of 20 baselines never do

FEASIBILITY

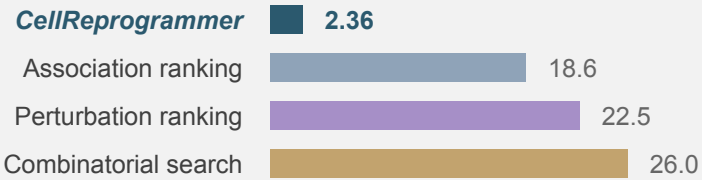
Targets with drug support



K = 3 · direction-matched LINCS support, not direct binding · category means

TOXICITY

Target-risk score



K = 3 · lower is safer · essential-gene risk · category means

TOXICITY

Downstream genes out of range



K = 3 · lower is better · outside the normal range · category means

CellReprogrammer: Duan et al., ICLR 2027 (under review).

Takeaways: From Data to Interventions

Each project answers one question along the way

Decoding disease

MUSE

CIKM 2025

What tissue patterns are shared across slices?

Builds one tissue map from many slices, even noisy ones.

KEY RESULT
Clearer tissue domains
accurate cross-slice alignment

iMiracle

CIKM 2024

How do neighboring cells influence gene activity?

Decodes which cell–cell signals control which genes.

KEY RESULT
Accurate LR–gene scores
and region-specific signals

Designing interventions

TargetSage

NeurIPS 2026

Which genes could be therapeutic targets?

Scores every gene as a target, with reasons a biologist can check.

KEY RESULT
Leads across 15 tasks
predicts future targets

CellReprogrammer

ICLR 2027 · under review

Which gene combinations shift cells toward a desired state?

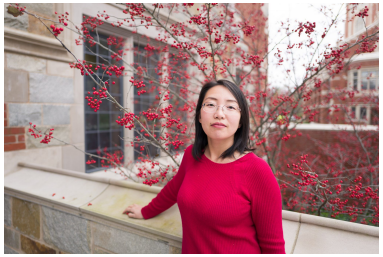
Selects small gene sets for predicted reversal, with drug & risk limits.

KEY RESULT
Strongest at every K
vs. 20 baselines · 5 disorders

Together, they turn biological data into testable hypotheses for new treatments.

Acknowledgments

The Zhang Lab · UC Irvine



Jing Zhang, Ph.D.
Associate Professor
Department of
Computer Science, UCI

Lab website





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NIH National Institute on Drug Abuse

Collaborators

SCORCH

M-SCORCH & Y-SCORCH group

